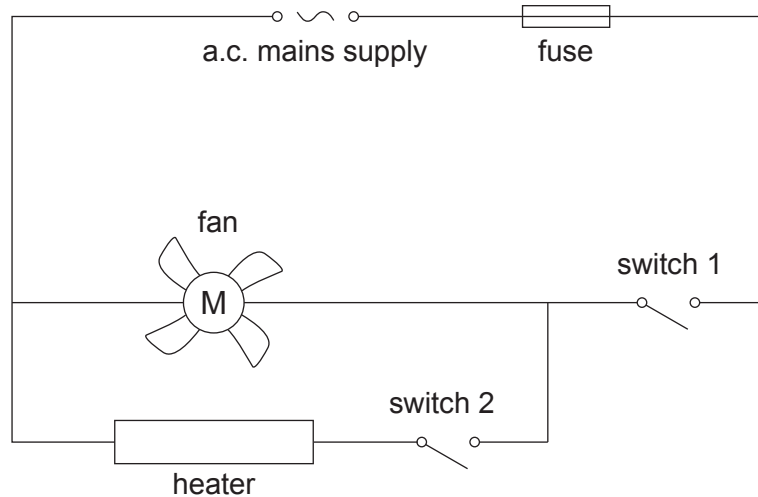


2. The circuit diagram below shows a design for a hairdryer that can blow either cold air or hot air.

The switches in the diagram are in the open position.



- (a) Complete the following sentences by underlining the correct phrase in the brackets. [2]

When switch 1 is closed and switch 2 is open the hairdryer blows cold air because

(**only the fan is on** / only the heater is on / the heater and fan are both on).

The hairdryer blows hot air when the heater and fan are both on. This happens when

(both switches are open / **only switch 2 is closed** / both switches are closed).

- (b) The hairdryer is used to blow hot air.

The current supplied is **6 A** and the mains voltage is 230 V.

- (i) Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the power of the hairdryer.

[2]

power = W



- (ii) The following fuses are available.

Tick (✓) **the box** to show the correct fuse that should be used for the hairdryer. [1]


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- (iii) Complete the following sentence by underlining the correct phrase in the brackets. [2]

Fuses switch off circuits if the

(**current is too high** / voltage is too high / power is too low)

to prevent (**electric shocks** / wires overheating / power cuts).

- (c) A **different hairdryer** is now used. The power of this hairdryer is **1.5 kW**.

- (i) Use the equation:

$$\text{units used (kWh)} = \text{power (kW)} \times \text{time (h)}$$

to calculate the units used (kWh) by this hairdryer in **2 hours**.

[2]

units used = kWh



- (ii) The cost per unit of electrical energy is 30 p.
Use your answer to part (c)(i) and the equation:

$$\text{cost} = \text{units used} \times \text{cost per unit}$$

to calculate the cost of using this hairdryer for 2 hours.

[2]

cost = p

- (iii) State the cost of using this hairdryer for **double** the time.

[1]

cost = p

- (iv) Tick (✓) **the box** that correctly shows the power of the 1.5 kW hairdryer in watts (W).

[1]

☐

1 500 000 W

☐

1 500 W

☐

1.5 W

☐

0.015 W

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